

BITS PILANI - WORK INTEGRATED LEARNING

PROGRAMMES DIVISION

M.TECH EMBEDDED SYSTEMS

ESCLZG517

**INTERNET OF THINGS (IOT) DESIGN AND
DEVELOPMENT**

SEMESTER 2

LAB ASSIGNMENT 1

STUDENT DETAILS:

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GITHUB REPOSITORY : <https://github.com/RaagulKarthiken/ESZG517-IoTLab>

L1 — Publisher Screenshot:

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[MQTT] Broker confirmed message
[01:47:43] PUB --> eszg517/lab/weather/2025ca01031
{"device_id": "2025ca01031", "temperature": 24.47, "humidity": 59.88, "pressure": 1013.1, "co2_ppm": 403.5, "aqi": 37.1, "light_lux": 7870.7, "wind_speed": 8.5, "wind_direction": 158.5, "rainfall_mm": 0.32, "battery_pct": 80.7, "timestamp": 1781381863}

[MQTT] Broker confirmed message
[01:47:48] PUB --> eszg517/lab/weather/2025ca01031
{"device_id": "2025ca01031", "temperature": 25.49, "humidity": 61.08, "pressure": 1013.92, "co2_ppm": 412.0, "aqi": 34.7, "light_lux": 7470.4, "wind_speed": 12.2, "wind_direction": 179.6, "rainfall_mm": 1.01, "battery_pct": 97.1, "timestamp": 1781381868}

[MQTT] Broker confirmed message
[01:47:53] PUB --> eszg517/lab/weather/2025ca01031
{"device_id": "2025ca01031", "temperature": 25.2, "humidity": 63.73, "pressure": 1013.68, "co2_ppm": 413.3, "aqi": 42.1, "light_lux": 6790.2, "wind_speed": 16.4, "wind_direction": 161.0, "rainfall_mm": 0.35, "battery_pct": 93.3, "timestamp": 1781381873}

[MQTT] Broker confirmed message
[01:47:58] PUB --> eszg517/lab/weather/2025ca01031
{"device_id": "2025ca01031", "temperature": 25.67, "humidity": 63.02, "pressure": 1013.78, "co2_ppm": 414.6, "aqi": 46.4, "light_lux": 9612.2, "wind_speed": 16.3, "wind_direction": 152.9, "rainfall_mm": 0, "battery_pct": 79.8, "timestamp": 1781381878}

[MQTT] Broker confirmed message
[01:48:03] PUB --> eszg517/lab/weather/2025ca01031
{"device_id": "2025ca01031", "temperature": 26.07, "humidity": 64.5, "pressure": 1013.14, "co2_ppm": 408.0, "aqi": 40.7, "light_lux": 8802.7, "wind_speed": 9.9, "wind_direction": 166.3, "rainfall_mm": 0.23, "battery_pct": 84.8, "timestamp": 1781381883}

[MQTT] Broker confirmed message
[01:48:08] PUB --> eszg517/lab/weather/2025ca01031
{"device_id": "2025ca01031", "temperature": 26.89, "humidity": 67.32, "pressure": 1012.86, "co2_ppm": 421.3, "aqi": 43.5, "light_lux": 6508.5, "wind_speed": 15.9, "wind_direction": 185.4, "rainfall_mm": 0.22, "battery_pct": 91.8, "timestamp": 1781381888}

[MQTT] Broker confirmed message
[01:48:13] PUB --> eszg517/lab/weather/2025ca01031
{"device_id": "2025ca01031", "temperature": 26.73, "humidity": 68.81, "pressure": 1013.65, "co2_ppm": 425.3, "aqi": 51.4, "light_lux": 3799.6, "wind_speed": 12.6, "wind_direction": 189.8, "rainfall_mm": 0, "battery_pct": 96.1, "timestamp": 1781381893}

[MQTT] Broker confirmed message
[01:48:18] PUB --> eszg517/lab/weather/2025ca01031
{"device_id": "2025ca01031", "temperature": 26.74, "humidity": 68.38, "pressure": 1013.29, "co2_ppm": 430.9, "aqi": 46.0, "light_lux": 8423.8, "wind_speed": 13.6, "wind_direction": 190.4, "rainfall_mm": 0, "battery_pct": 84.5, "timestamp": 1781381898}
```

L2 — Subscriber Screenshot:

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[DATA] 2025ca01031 | temperature=25.2°C humidity=63.73% pressure=1013.68Pa co2=413.3ppm aqi=42.1 light=6790.2lux windspeed=16.4Kph direction=161.0 rainfall=0.35mm battery_pct=93.3 dBm
[INFLUX] Written: 2025ca01031 temp=25.2 hum=63.73 pres=1013.68 co2=413.3 aqi=42.1 light=6790.2 wind=16.4 dir=161.0 rain=0.35 bat=93.3
[DATA] 2025ca01031 | temperature=25.67°C humidity=63.02% pressure=1013.78Pa co2=414.6ppm aqi=46.4 light=9612.2lux windspeed=16.3Kph direction=152.9 rainfall=0mm battery_pct=79.8 dBm
[INFLUX] Written: 2025ca01031 temp=25.67 hum=63.02 pres=1013.78 co2=414.6 aqi=46.4 light=9612.2 wind=16.3 dir=152.9 rain=0 bat=79.8
[DATA] 2025ca01031 | temperature=26.07°C humidity=64.5% pressure=1013.14Pa co2=408.0ppm aqi=40.7 light=8802.7lux windspeed=9.9Kph direction=166.3 rainfall=0.23mm battery_pct=84.8 dBm
[INFLUX] Written: 2025ca01031 temp=26.07 hum=64.5 pres=1013.14 co2=408.0 aqi=40.7 light=8802.7 wind=9.9 dir=166.3 rain=0.23 bat=84.8
[DATA] 2025ca01031 | temperature=26.89°C humidity=67.32% pressure=1012.86Pa co2=421.3ppm aqi=43.5 light=6508.5lux windspeed=15.9Kph direction=185.4 rainfall=0.22mm battery_pct=91.8 dBm
[INFLUX] Written: 2025ca01031 temp=26.89 hum=67.32 pres=1012.86 co2=421.3 aqi=43.5 light=6508.5 wind=15.9 dir=185.4 rain=0.22 bat=91.8
[DATA] 2025ca01031 | temperature=26.73°C humidity=68.81% pressure=1013.65Pa co2=425.3ppm aqi=51.4 light=3799.6lux windspeed=12.6Kph direction=189.8 rainfall=0mm battery_pct=96.1 dBm
[INFLUX] Written: 2025ca01031 temp=26.73 hum=68.81 pres=1013.65 co2=425.3 aqi=51.4 light=3799.6 wind=12.6 dir=189.8 rain=0 bat=96.1
[DATA] 2025ca01031 | temperature=26.74°C humidity=68.38% pressure=1013.29Pa co2=430.9ppm aqi=46.0 light=8423.8lux windspeed=13.6Kph direction=190.4 rainfall=0mm battery_pct=84.5 dBm
[INFLUX] Written: 2025ca01031 temp=26.74 hum=68.38 pres=1013.29 co2=430.9 aqi=46.0 light=8423.8 wind=13.6 dir=190.4 rain=0 bat=84.5
[DATA] 2025ca01031 | temperature=27.59°C humidity=70.12% pressure=1013.24Pa co2=411.4ppm aqi=49.1 light=6032.8lux windspeed=20.3Kph direction=180.2 rainfall=0.15mm battery_pct=96.4 dBm
[INFLUX] Written: 2025ca01031 temp=27.59 hum=70.12 pres=1013.24 co2=411.4 aqi=49.1 light=6032.8 wind=20.3 dir=180.2 rain=0.15 bat=96.4
[DATA] 2025ca01031 | temperature=27.8°C humidity=71.43% pressure=1013.58Pa co2=439.1ppm aqi=55.2 light=5298.4lux windspeed=11.9Kph direction=198.4 rainfall=0.34mm battery_pct=89.7 dBm
[INFLUX] Written: 2025ca01031 temp=27.8 hum=71.43 pres=1013.58 co2=439.1 aqi=55.2 light=5298.4 wind=11.9 dir=198.4 rain=0.34 bat=89.7
[DATA] 2025ca01031 | temperature=27.72°C humidity=71.43% pressure=1013.96Pa co2=424.9ppm aqi=54.2 light=7634.8lux windspeed=20.1Kph direction=200.7 rainfall=0.06mm battery_pct=78.5 dBm
[INFLUX] Written: 2025ca01031 temp=27.72 hum=71.43 pres=1013.96 co2=424.9 aqi=54.2 light=7634.8 wind=20.1 dir=200.7 rain=0.06 bat=78.5
[DATA] 2025ca01031 | temperature=28.3°C humidity=73.79% pressure=1013.53Pa co2=431.0ppm aqi=48.8 light=8400.4lux windspeed=22.1Kph direction=190.0 rainfall=0.35mm battery_pct=79.3 dBm
[INFLUX] Written: 2025ca01031 temp=28.3 hum=73.79 pres=1013.53 co2=431.0 aqi=48.8 light=8400.4 wind=22.1 dir=190.0 rain=0.35 bat=79.3
[DATA] 2025ca01031 | temperature=28.39°C humidity=71.55% pressure=1012.64Pa co2=429.3ppm aqi=51.4 light=5642.9lux windspeed=21.4Kph direction=206.0 rainfall=0mm battery_pct=86.8 dBm
[INFLUX] Written: 2025ca01031 temp=28.39 hum=71.55 pres=1012.64 co2=429.3 aqi=51.4 light=5642.9 wind=21.4 dir=206.0 rain=0 bat=86.8
[DATA] 2025ca01031 | temperature=28.5°C humidity=73.93% pressure=1013.24Pa co2=449.3ppm aqi=53.8 light=7648.1lux windspeed=25Kph direction=204.4 rainfall=0.66mm battery_pct=91.2 dBm
[INFLUX] Written: 2025ca01031 temp=28.5 hum=73.93 pres=1013.24 co2=449.3 aqi=53.8 light=7648.1 wind=25 dir=204.4 rain=0.66 bat=91.2
[DATA] 2025ca01031 | temperature=28.79°C humidity=73.94% pressure=1013.39Pa co2=439.4ppm aqi=57.5 light=6132.2lux windspeed=23.6Kph direction=199.6 rainfall=0mm battery_pct=78.8 dBm
[INFLUX] Written: 2025ca01031 temp=28.79 hum=73.94 pres=1013.39 co2=439.4 aqi=57.5 light=6132.2 wind=23.6 dir=199.6 rain=0 bat=78.8
[DATA] 2025ca01031 | temperature=28.67°C humidity=73.4% pressure=1013.74Pa co2=443.4ppm aqi=50.0 light=8235.6lux windspeed=22.1Kph direction=216.5 rainfall=0.23mm battery_pct=83.5 dBm
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L2 — Grafana Dashboard Screenshot:



THANK YOU